

IT3091 – MACHINE LEARNING

Introduction to Data Mining and Machine Learning

Definitions & Importance • The Data Mining Process • Key Concepts & Terminology

Topic 1 – Introduction to Data Mining and Machine Learning

Today's Roadmap



Definitions & Importance

What is data mining? What is machine learning? Why do they matter, and where are they used?



The Data Mining Process

How raw data becomes knowledge — the KDD process and the CRISP-DM methodology.



Key Concepts & Terminology

The vocabulary you'll use all semester: datasets, features, learning paradigms, model-building terms.

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Definitions & Importance

What data mining and machine learning are — and why they matter

What is Data Mining?

The process of discovering non-trivial, previously unknown, and potentially useful patterns, relationships, and structure hidden in large volumes of data.

- Sits at the intersection of statistics, databases, and computing — born from the need to make sense of growing organizational data in the 1990s.
- Emphasis is on knowledge discovery: finding patterns a human analyst did not think to look for.
- Operates on data that already exists — typically stored in databases, warehouses, or data lakes.
- The output is insight for human decision-makers, not necessarily an automated action.

A BIT OF HISTORY

The term “data mining” became popular in the 1990s as organizations began storing far more data than they could analyze manually.

It is closely linked to the broader field of Knowledge Discovery in Databases (KDD) — data mining is technically the pattern-extraction step within the larger KDD process (covering in Part B).

“Mining” is a metaphor: like mining ore from rock, valuable knowledge is extracted from raw data.

What is Machine Learning?

A field of computing in which systems improve their performance on a task through experience (data), rather than through explicitly programmed rules.

- Classic framing (Tom Mitchell): a program learns from experience E on task T with performance P if its performance on T , measured by P , improves with E .
- Instead of hand-coding rules, an algorithm infers a model from labelled or unlabelled examples.
- The output is typically a model that generalizes to new, unseen data — and is used to automate a decision or prediction.
- Rooted in computer science and statistics, and closely tied to artificial intelligence.

RULE-BASED vs. LEARNED

Traditional programming

Rules + Data $\rightarrow$ written by a human $\rightarrow$ Output

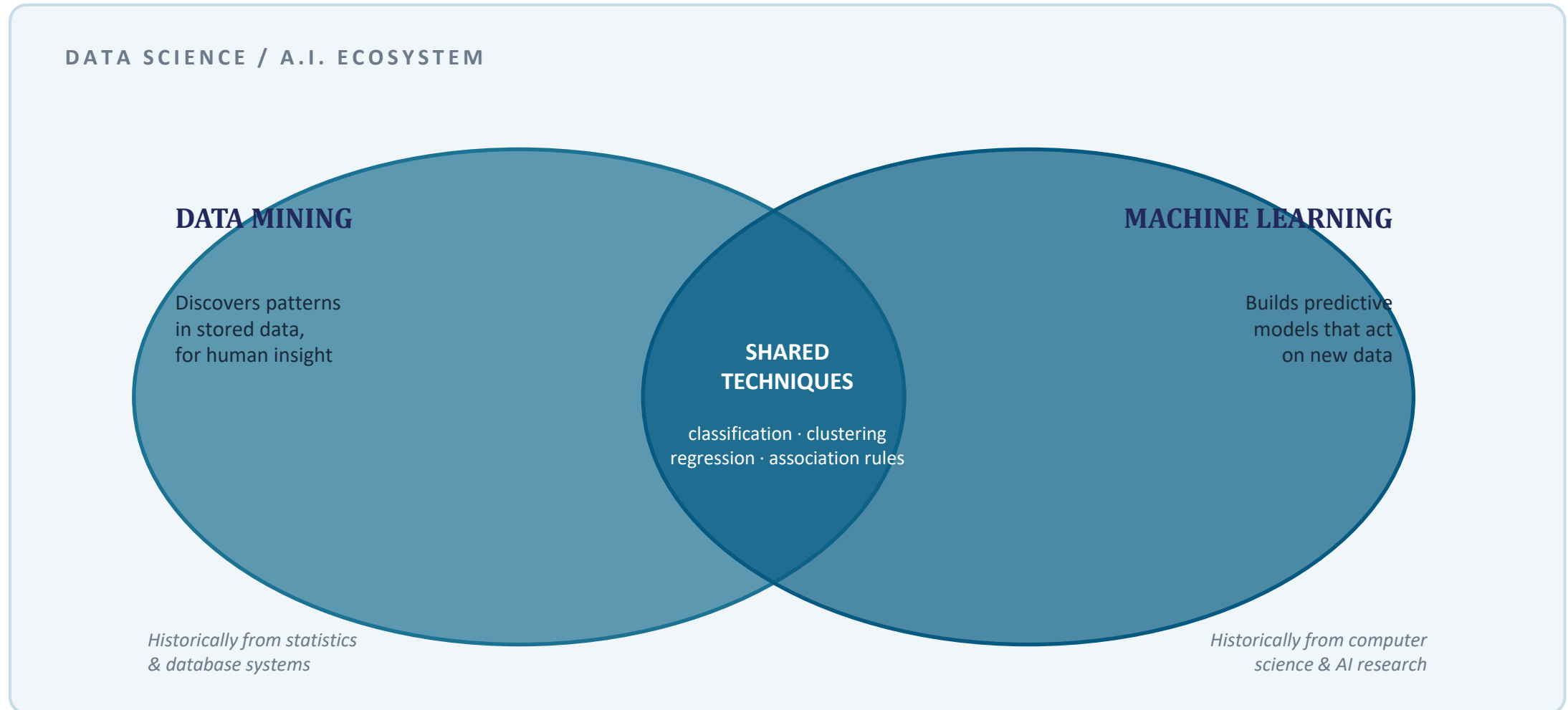
Machine learning

Data + Output examples $\rightarrow$ algorithm infers $\rightarrow$ Rules (the model)

This is why ML shines on problems too complex to hand-code: spam detection, image recognition, fraud detection.

How They Relate

Data mining and machine learning overlap heavily, but they are not identical



Why This Matters

Data is growing far faster than our ability to look at it by hand

180+ ZB

of data expected to be generated globally
in 2025

2.5 QT

bytes of data created every day
(quintillion)

90%

of the world's data was created in just
the last few years

- Humans cannot manually inspect millions of rows — automated pattern discovery is the only scalable option.
- Competitive advantage: organizations that turn data into decisions outperform those that don't.
- Enables personalization and automation at a scale no manual process could match (recommendations, fraud alerts, diagnosis support).
- A core, in-demand skill set for computing graduates across almost every industry.

Real-World Applications

Data mining and ML now underpin decisions across almost every sector



Healthcare

Disease risk prediction, medical image analysis, treatment recommendation



Finance

Credit scoring, algorithmic trading, fraud & anomaly detection



Retail & E-commerce

Recommendation engines, demand forecasting, market-basket analysis



Cybersecurity

Intrusion detection, malware classification, spam filtering



Social Media

Content ranking, sentiment analysis, targeted advertising



Manufacturing

Predictive maintenance, quality control, process optimization

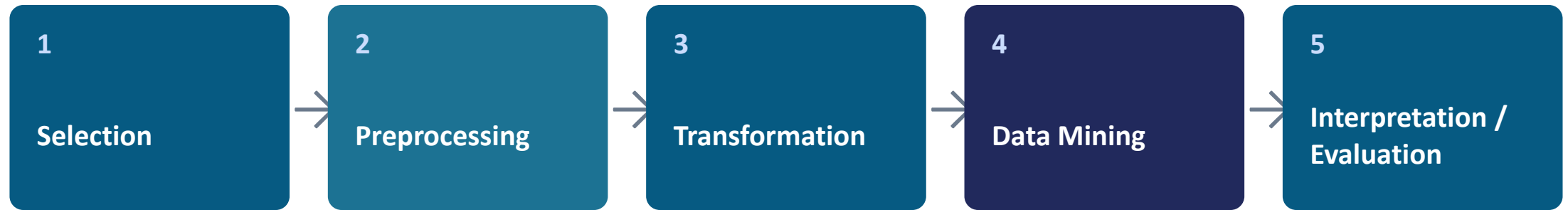
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The Data Mining Process

How raw data becomes usable knowledge — step by step

The KDD Process

Knowledge Discovery in Databases – data mining is only one step of a larger pipeline (Fayyad et al., 1996)

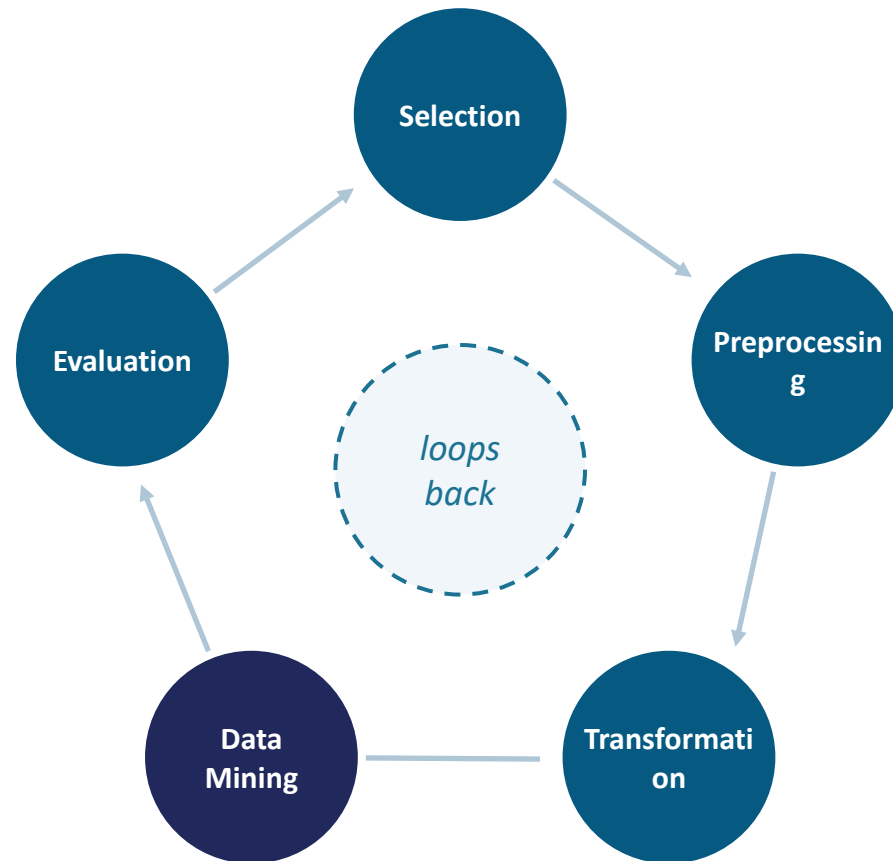


Target Data → Preprocessed Data → Transformed Data → Patterns → Knowledge

- 1. Selection:** Identify and pull the relevant subset of data from larger sources / databases.
- 2. Preprocessing:** Clean the data — handle noise, missing values, and inconsistencies.
- 3. Transformation:** Convert data into a form suitable for mining (aggregation, normalization, feature construction).
- 4. Data Mining:** Apply algorithms to extract patterns — classification, clustering, association rules, etc.
- 5. Interpretation / Evaluation:** Assess and visualize the discovered patterns to identify genuinely useful knowledge.

KDD Is Iterative, Not Linear

In practice, analysts loop back through earlier stages as they learn more about the data

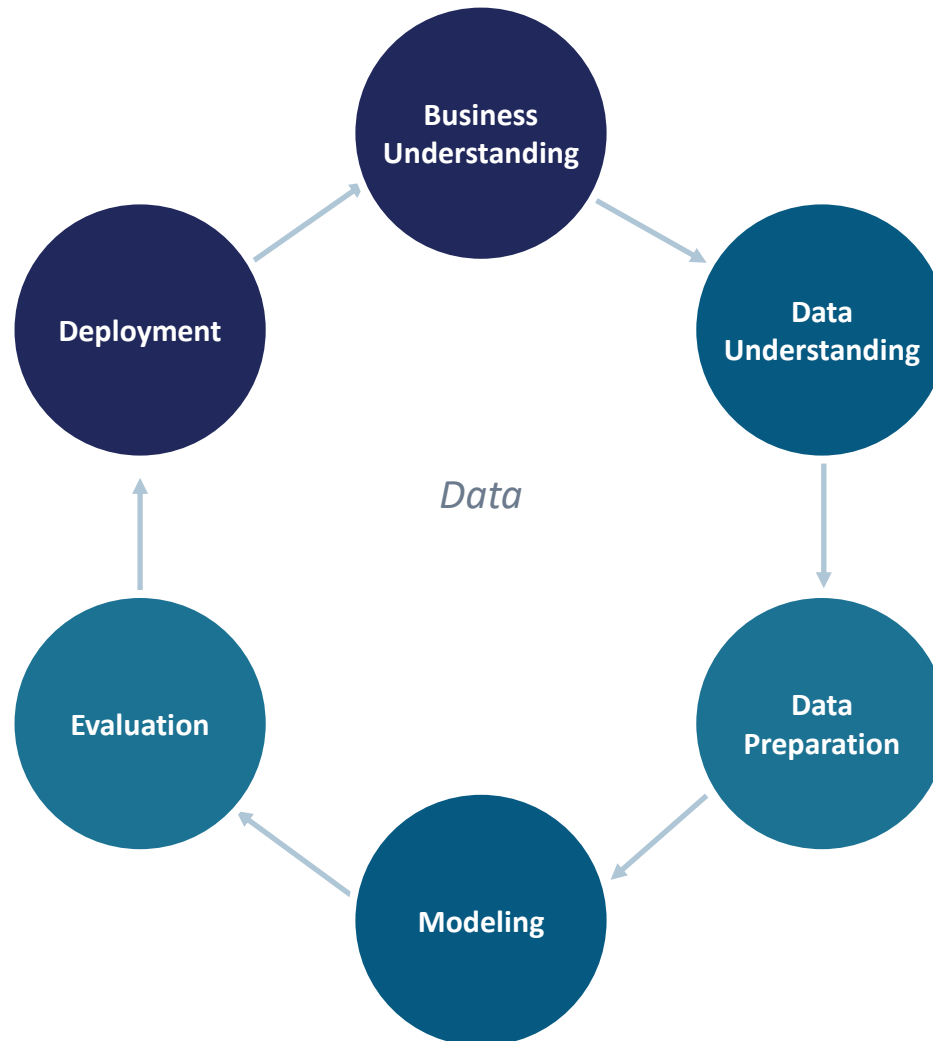


WHY THE LOOPS HAPPEN

- Cleaning often reveals that more raw data needs to be selected.
- Mining results may be weak — prompting a return to transformation for better features.
- Evaluation may show patterns are trivial or already known, sending the analyst back to try a different technique.
- Real datasets are messy: understanding deepens only after you've tried mining once.

CRISP-DM Methodology

Cross-Industry Standard Process for Data Mining – today's most widely used industry methodology



KEY IDEAS

- Industry-neutral: usable regardless of sector or tool.
- Starts with business goals — not the data — so mining stays aligned with a real question.
- Cyclic: deployment often raises new business questions, restarting the cycle.
- Widely taught alongside KDD; CRISP-DM adds explicit business framing and deployment.

KDD vs. CRISP-DM

Two complementary views of the same underlying process

Dimension	KDD Process	CRISP-DM
Origin	Academic research (Fayyad et al., 1996)	Industry consortium (1996-2000, cross-vendor)
Starting point	Target data selection	Business understanding / objectives
Number of stages	5 stages	6 phases
Explicit business framing	Implicit	Explicit — first phase
Deployment as a phase	Not explicit (folded into interpretation)	Explicit final phase
Typical use today	Framing academic / conceptual discussion	Framing real-world / industry projects



C

Key Concepts & Terminology

The shared vocabulary you will use for the rest of the semester

Core Terms: Describing Data

Every dataset can be described using this small, consistent vocabulary

CustomerID	Age	Income	Churn?
1024	34	52,000	No
1025	51	88,500	Yes
1026	29	31,200	No

Dataset = the whole table • Instance/Record = one row • Attribute/Feature = one column • Label = the target column ("Churn?")

Dataset: A structured collection of data used for analysis or model building.

Instance / Record / Observation: A single row — one data point (e.g., one customer).

Attribute / Feature / Variable: A single column — one measured property (e.g., Age).

Label / Target / Class: The attribute you are trying to predict or explain (only present in supervised problems).

Types of Data

The type of data available shapes which techniques you can apply



Structured

Organized into rows & columns with a fixed schema.

EXAMPLES

Spreadsheets, relational database tables, sensor logs

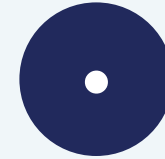


Semi-Structured

Has some organizational structure but not a fixed schema.

EXAMPLES

JSON, XML, emails with headers



Unstructured

No predefined structure at all.

EXAMPLES

Free text, images, audio, video

Roughly 80% of organizational data is estimated to be unstructured or semi-structured — a major driver of modern ML adoption (deep learning especially).

Learning Paradigms

How an algorithm learns depends on what kind of feedback it receives

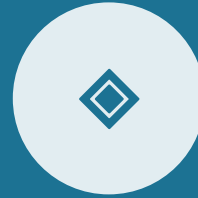


Supervised Learning

Learns from labelled examples (input → correct output) to predict labels for new data.

EXAMPLE

Predicting house prices; classifying email as spam/not spam



Unsupervised Learning

Learns from unlabelled data, finding structure or groupings on its own.

EXAMPLE

Customer segmentation; market-basket analysis



Reinforcement Learning

Learns by taking actions in an environment and receiving reward/penalty feedback.

EXAMPLE

Game-playing agents; robotics; recommendation ranking

Model-Building Vocabulary

The terms you'll hear in every lab session from here on

FULL DATASET SPLIT

Training Set

~70%

Validation Set

~15%

Test Set

~15%

Training Set: Data the algorithm learns from.

Validation Set: Held-out data used to tune the model during development.

Test Set: Untouched data used once, at the end, to judge final performance.

Model: The learned representation (e.g., equation, tree, set of weights) that maps inputs to outputs.

Algorithm: The learning procedure used to produce the model from training data (e.g., decision tree induction).

Parameters: Values the algorithm learns automatically from data (internal to the model).

Hyperparameters: Settings chosen before training that control how learning happens (e.g., tree depth).

Inference / Prediction: Using a trained model to produce an output for new, unseen input.

Common Pitfalls to Know By Name

Two terms you will hear constantly for the rest of the module

Underfitting

The model is too simple to capture the real pattern in the data.

Poor performance on training AND new data.

Good Fit

The model captures the true underlying pattern without chasing noise.

Strong, stable performance on new data.

Overfitting

The model memorizes noise / quirks specific to the training data.

Great on training data, poor on new data.

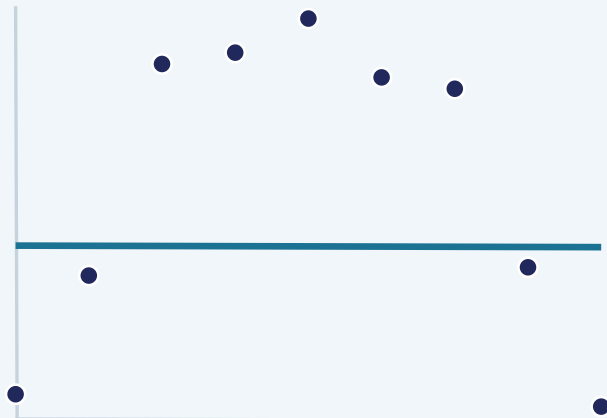
THE BIAS-VARIANCE TRADEOFF

Underfitting is a symptom of high bias (overly simplistic assumptions). Overfitting is a symptom of high variance (too sensitive to the specific training sample). Model evaluation — covered later in the module — is largely about finding the sweet spot between the two.

Visualizing the Fit

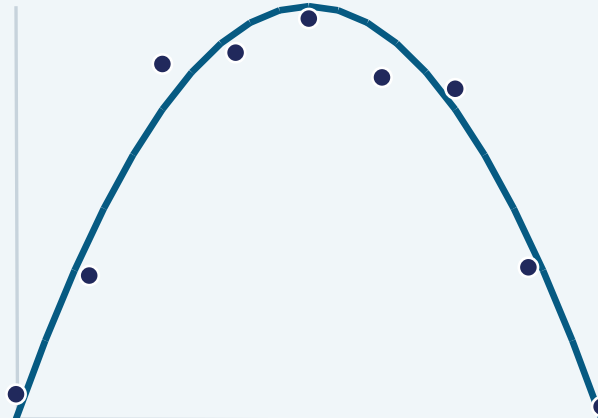
The same dataset, fitted three different ways

Underfitting · High Bias



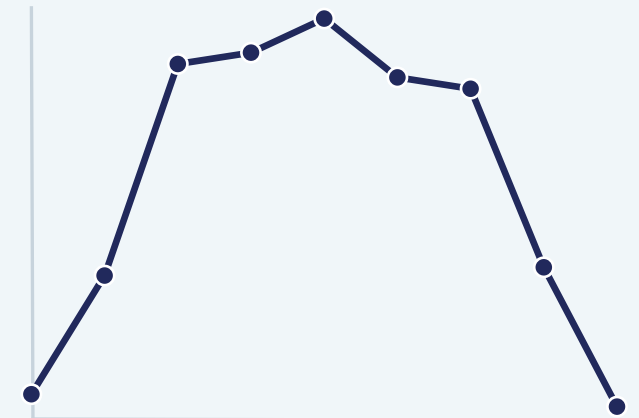
A straight line ignores the hump entirely — misses the pattern in both training and new data.

Good Fit · Balanced



A smooth curve follows the true trend and treats the scatter around it as noise.

Overfitting · High Variance



The line bends to touch every point exactly — it has memorized the noise, not the pattern.

• data points (with noise) — fitted model line / curve

Key Takeaways

A

Definitions & Importance

Data mining discovers patterns for human insight; ML builds models that act on new data. Both matter because data now vastly outpaces manual analysis.

B

The Data Mining Process

Raw data becomes knowledge through a structured, iterative process — KDD's 5 stages, or CRISP-DM's 6 business-first phases.

C

Key Concepts & Terminology

Datasets, features, labels, learning paradigms, train/validation/test splits, and overfitting/underfitting form the vocabulary for the rest of the semester.

Looking Ahead

INDEPENDENT STUDY / PREP

- Revise Python and pandas basics (from IT2011).
- Revisit basic ML terminology introduced in IT2011.
- Skim Ch.1 of Géron's "Hands-On Machine Learning" for a practitioner's view of the field.
- Skim Ch.1 of Trappenberg's "Fundamentals of Machine Learning" for the conceptual foundations covered today.

Expected output: a short problem-framing note + a baseline notebook, in preparation for the next lab session.

NEXT LECTURE

Data Preprocessing

- Data cleaning methods
- Data integration and transformation
- Feature selection & dimensionality reduction
- Handling imbalanced datasets

Bring: the mind-map / note you sketch of a messy real-world data scenario — we will use it to frame next week's hands-on preprocessing work.

Questions?

Thank you – see you next lecture for Data Preprocessing
